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Electronics and Communications Technology

Electronic Devices and Circuit Design

Challenges and Applications in the Internet of Things

Editors: Suman Lata Tripathi, PhD
Smrity Dwivedi, PhD

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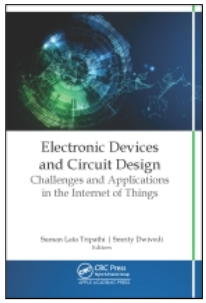

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This new volume, **Electronic Devices and Circuit Design: Challenges and Applications in the Internet of Things**, offers a broad view of the challeges of electronic devices and circuits for IoT applications in a concise way.. The book presents the basic concepts and fundamentals behind new low power, high-speed efficient devices, circuits, and systems with new technology in addition to CMOS. It aims to help develop an understanding of new materials to improve device performance with smaller dimensions and lower costs. It looks at the new methodologies to enhance system performance and provides key parameters to explore the devices and circuit performance based on smart applications. The volume bridges the gap for researchers working on different areas of smart devices, circuits, and systems with IoT applications.

The fundamental principles of electronic device and circuit design are discussed in a clear and detailed manner along with explanatory diagrams. Various IoT-based smart applications in different fields along with challenges and issues along with prospects of IoT for future applications have also been discussed.

The chapters delve into myriad aspects of circuit design, including MOSFET structures depending on their low power applications for IoT-enabled systems, advanced sensor design and fabrication using MEMS, indirect bootstrap techniques, efficient CMOS comparators, various encryption-decryption algorithms, IoT video forensics applications, microstrip patch antennas in embedded IoT applications, real-time object detection using sound, IOT and nanotechnologies based wireless sensors, and much more.

It also introduces logic BIST using BS-CFSR for electronic devices that can be part of many applications in many industries, such as defense, automotive, banking, computer, healthcare, networking, and telecommunication.

This book provides a treasure house of up-to-date information on many aspects of electronic devices and circuit design, making it highly beneficial for researchers, scientists, faculty, students, and industry professionals.

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Suman Lata Tripathi, PhD, is associated with Lovely Professional University as a Professor with more than 17 years of experience in academics. She has published more than 45 research papers in refereed journals and conferences and holds several patents. She has organized workshops, summer internships,and expert lectures for students. She has worked as a session chair, conference steering committee member, editorial board member, and reviewer in international and national IEEE journals and conferences. She has been awarded a Research Excellence Award in 2019 at Lovely Professional University and has received the best paper award at IEEE ICICS-2018. She is also associated as an editor/author of books and book series with publishers such as CRC Press, Taylor & Francis, Springer, Nova Science Publishers, Scrivener Publishing Wiley, IGI Global Publisher, Apple Academic Press, etc. Her area of expertise includes microelectronics device modeling and characterization, low power VLSI circuit design, VLSI design of testing and advance FET design for IoT and biomedical applications, etc. She also has an interest in the area of the energy-efficient systems, green energy, and renewable energy. She has completed her PhD in the area of microelectronics and VLSI from MNNIT, Allahabad, India; her MTech in Electronics Engineering from UP Technical University, Lucknow; and BTech in Electrical Engineering from Purvanchal University, Jaunpur, India.

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